

Order Distribution

1. Overview

The **Order Distribution System** is responsible for automatically distributing orders from the **Claimable Pool** to available **TeleSales Agents**.

The system should distribute orders fairly among eligible agents based on:

- Agent availability/status
- Number of currently pending orders
- Most recently assigned order time
- Maximum pending order limit per agent

The objective is to ensure that orders are distributed as evenly as possible among available agents while preventing any agent from receiving more than **10 Currently Pending Orders**.

2. Key Concepts

Term	Definition
Claimable Pool	The pool where orders are available for distribution to TeleSales Agents.
Currently Pending	Orders currently assigned to an agent but not yet completed/confirmed.
Online Agent	An agent who is currently available to receive orders.
Offline Agent	An agent who is not available to receive new orders.
Maximum Pending Limit	An agent can have a maximum of 10 Currently Pending Orders .
Recent Assignment	The agent who most recently received an order from the Claimable Pool.
Least Recently Assigned	Among eligible agents, the agent who has waited the longest since

receiving an order.

3. Eligible Agent Criteria

An agent is eligible to receive an order from the Claimable Pool when:

1. The agent is **Online**.
2. The agent has **less than 10 Currently Pending Orders**.
3. The agent is active and available for order distribution.

An agent should **not** receive a new order when:

- The agent is Offline.
- The agent already has **10 Currently Pending Orders**.

4. Basic Distribution Logic

When a new order becomes available in the **Claimable Pool**, the system identifies all eligible agents.

The order should be assigned to the eligible agent who is **least recently assigned an order**.

This ensures that an agent who has not received an order recently gets priority over an agent who has just received an order.

Example

Assume there are three Online Agents:

- Agent 1
- Agent 2
- Agent 3

Initially:

Agent	Currently Pending
Agent 1	0
Agent 2	0
Agent 3	0

Order 1

The first order is distributed to **Agent 1**.

Agent	Pending
Agent 1	1
Agent 2	0
Agent 3	0

Order 2

The next order is distributed to **Agent 2**.

Agent	Pending
Agent 1	1
Agent 2	1
Agent 3	0

Order 3

The next order is distributed to **Agent 3**.

Agent	Pending
Agent 1	1
Agent 2	1
Agent 3	1

Order 4

Agent 1 was the first to receive an order and has been waiting the longest.

Therefore, the next order goes to **Agent 1**. The same distribution principle continues.

5. Distribution Based on Recent Assignment

The system should not simply distribute orders based on a fixed sequence.

Instead, it should consider which eligible agent received an order most recently.

The agent who has **not received an order for the longest time** should receive the next available order.

For example:

Agent	Pending Orders	Last Received Order
Agent 1	6	Recent
Agent 2	4	Older
Agent 3	2	Oldest

The next order should go to **Agent 3** because Agent 3 has received an order less recently than the other eligible agents.

After receiving the order:

Agent	Pending Orders
Agent 1	6
Agent 2	4
Agent 3	3

If another order becomes available and Agent 3 is still the least recently assigned eligible agent, the next order may again go to Agent 3.

For example:

Agent	Before	After
Agent 1	6	6
Agent 2	4	4
Agent 3	3	4

Now the distribution should consider **assignment recency** again and prioritize the agent who has waited the longest.

6. Example of Equal Distribution

Agent	Pending
Agent 1	6

Agent 2	6
Agent 3	6

All three agents are Online and eligible.

The next order should be distributed according to the **assignment recency**.

If Agent 1 has waited the longest since receiving an order:

Next order → Agent 1

Result:

Agent	Pending
Agent 1	7
Agent 2	6
Agent 3	6

The system should then continue using the same recency-based distribution logic.

7. Offline Agent Handling

Agent availability must be considered before distributing every order.

An **Offline** Agent must be excluded from the distribution pool.

Example

Current state:

Agent	Pending	Status
Agent 1	7	Online
Agent 2	6	Online
Agent 3	6	Offline

If Agent 3 is **Offline**, the next order must **not** be assigned to Agent 3.

The system should distribute the order among the remaining eligible agents.

Therefore:

Next order → Agent 2

Result:

Agent	Pending	Status
Agent 1	7	Online
Agent 2	7	Online
Agent 3	6	Offline

8. Maximum Pending Order Limit

Each TeleSales Agent can receive a maximum of **10 Currently Pending Orders** from the Claimable Pool.

Example

Agent	Pending	Status	Eligible?
Agent 1	10	Online	No
Agent 2	8	Online	Yes
Agent 3	6	Online	Yes

Agent 1 must not receive another order because the maximum limit has already been reached.

The next order should be distributed between Agent 2 and Agent 3 according to the distribution/recency logic.

9. When an Agent Completes an Order

When an agent completes/confirms an order and the order is removed from Currently Pending, the agent becomes eligible to receive another order if they are Online.

Example:

Agent	Pending	Eligible?
Agent 1	10	No
Agent 2	8	Yes
Agent 3	7	Yes

Agent 1 completes one order.

New state:

Agent	Pending	Eligible?
Agent 1	9	Yes
Agent 2	8	Yes
Agent 3	7	Yes